Accurate on Average, Blind at the Fork

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Abstract: Before a robot executes an action, it should know whether its own uncertainty could change the outcome. People make this judgment with an intuitive physics engine [1]: they run a few noisy simulations over an uncertain scene and call it precarious when the outcomes disagree. We build this check for robots as a goal-free *fork probe*: sample the robot's measured perception and execution error, roll the intended action chunk forward 64 times, and flag it when the futures split. With a physics engine and a real camera-based perception model, the probe recovers fragile states on unseen Jenga layouts (κ up to 0.81, against 0.42 when perception uncertainty is ignored). Without the engine the probe needs a world model, and there it fails: a token-space world model that predicts 3–4$\times$ better than a no-motion baseline reaches only κ 0.19, because its sampled uncertainty has the right size but not where futures branch. Safe deployment needs world models with *branching fidelity*, which average-error benchmarks do not measure.

Keywords: World models, intuitive physics, uncertainty, safe deployment

1 Introduction

Manipulation policies, from diffusion policies to vision-language-action models, act by committing to short action chunks [2, 3, 4]. Deploying them safely requires a check before each chunk runs. Existing monitors address *epistemic* uncertainty, flagging inputs the model has not seen [5, 6]. We address a second kind that remains even for a perfect model: *physical* uncertainty. In contact-rich tasks such as Jenga extraction, the same command can slide a block out or tip a neighbour depending on millimetres of execution error and perception error. Such a chunk is precarious even if its nominal outcome is fine, because uncertainty the robot cannot control decides what happens.

People handle this with an *intuitive physics engine* (IPE): a few approximate simulations over an uncertain scene, whose disagreement signals precariousness [1, 7, 8]. Our goal is to give robots the same ability, which needs two things: an honest model of the robot's own uncertainty, and a model to simulate with. Outside simulation the second must be learned, which makes the debate on whether robots need world models concrete: a world model is useful here only if its rollouts branch where the real dynamics branch, a property prediction error does not measure.

Contributions. (1) A *fork probe*: a pre-execution check that propagates the robot's measured perception and execution uncertainty through a rollout engine and flags splits, with no task labels, failure data or fitted thresholds (Sec. 2). (2) Validation with a physics engine and real camera perception on three unseen test sets, which sets the reference a world model must reach (Sec. 3.1). (3) *Branching fidelity* as a world-model evaluation, and two learned token-space world models that predict well on average yet fail it, with a diagnosis (Sec. 3.2).

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2 Setup and the fork probe

Task. A simulated Franka Panda (MuJoCo) extracts blocks from Jenga-like stacks. At each chunk start a policy commits to an 8-step action chunk; ours come from recorded episodes, but the probe treats the policy as a black box and sees only the chunk, one camera image and proprioception.

Fragility. A chunk is *fragile* if the robot's uncertainty, its execution error and its error in perceiving the scene, can send the outcome into qualitatively different regimes. This needs no goal or failure label, which a monitor around a generalist policy cannot assume, and it isolates exactly the cases where intent no longer determines the outcome. For our task we measure it physically: a state is a *fork* if some but not all of 64 futures sampled from this uncertainty (2–62) topple a neighbouring block. A chunk that fails every time is not a fork and needs a task-level check.

Uncertainty model. Execution error comes from the robot's measured tracking residuals (median chunk maximum 3.2 mm). Perception is a frozen DINOv2 encoder [9] on one close-camera image with a keypoint head for the three blocks in play (2.5 mm held-out error); its error model is measured once, offline, as a zero-mean Gaussian fitted to its errors on held-out images with known poses (a full 9×9 position covariance capturing correlated errors, plus rotation covariances).

Fork probe and rollout engine (Fig. 1a). The probe imagines 64 versions of the chunk, each starting from the perceived scene plus one perception-error draw and executing the chunk minus one execution-error snippet, followed by a 30-step imagined hold so the outcome settles. It flags the chunk if, at hold steps 10 and 30, the endings (object positions or image tokens) projected on their first principal axis prefer two Gaussians to one by BIC, are well separated (Ashman's $D > 2$ [10]), have at least 2 members in the smaller group, and keep the same grouping; these are fixed statistical conventions, not fitted to labels. A variant draws 5 scene hypotheses and flags if at least 3 split. The probe works with any rollout engine. Our reference is MuJoCo with the three blocks placed at the sampled poses and everything else (geometry, masses, other blocks) taken from the simulator, which a real robot does not have; the learned alternative is a world model (Sec. 3.2).

Ground truth. One real outcome cannot grade a monitor, since a fragile chunk may succeed by luck. An oracle therefore samples the 64 futures from the *true* scene in exact physics and labels the state a fork as above. Two independent oracle runs agree at Cohen's κ 0.99, the ceiling. Each evaluation uses an unseen test set of stack layouts (up to 60 topple-screened states plus 100 safe ones) under a protocol frozen before the data existed.

3 Results

3.1 Propagating uncertainty through a physics engine works

With exact physics and the true scene, the split test flags all 23 topple forks among 112 development states, with 10% alarms on the 89 quiet ones. With real perception, perception uncertainty must be propagated: errors of 2–4 mm are comparable to execution noise, and a probe that trusts the estimate reaches κ 0.42–0.54, while the fork probe reaches 0.70–0.81 and catches 77–91% of forks (Table 1). Reading outcomes from DINOv2 tokens of rendered endings instead of object positions costs almost nothing, so the probe works from images alone: on test set C, κ 0.81 with 88% of forks caught (95% CI 0.77–0.94) and 7% false alarms. This is the reference a world model must reach.

3.2 Propagating uncertainty through learned world models fails

Without privileged geometry a learned model must take MuJoCo's role, and the probe becomes its test: given one image and the 64 perturbed chunks, do its rollouts branch where physics branches? Since outcomes can be read from image tokens, a token-space world model can be probed like the simulator. Our model compresses DINOv2 tokens by a fixed per-patch PCA to a $28 \times 28 \times 8$ latent

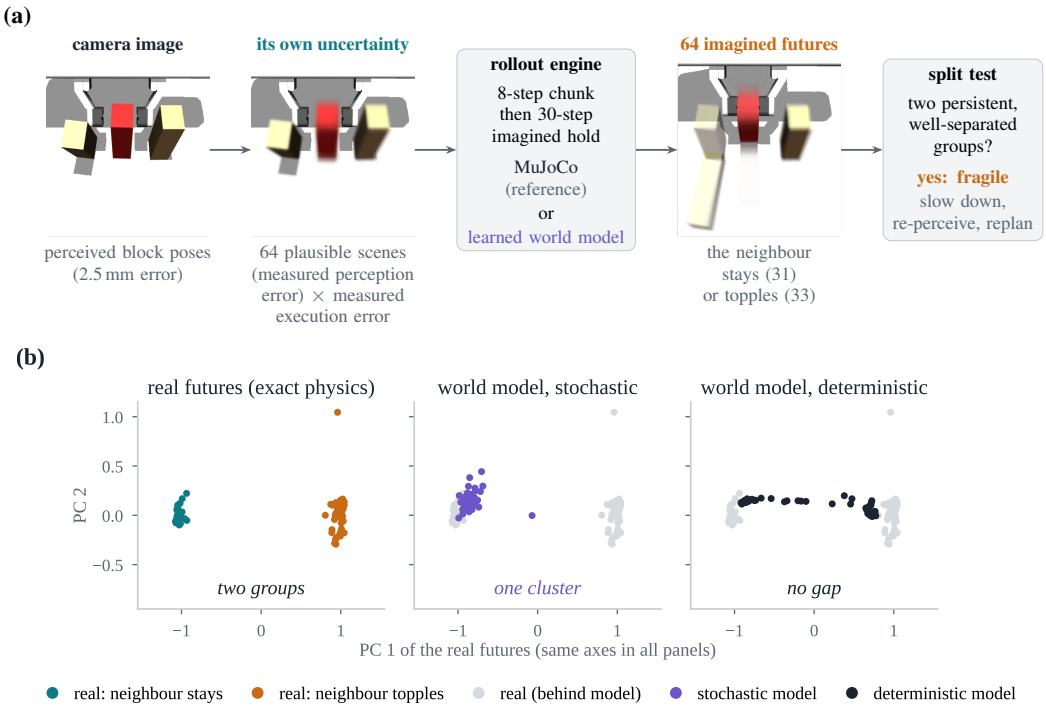


Figure 1: (a) The fork probe on one held-out state (exact renders; the uncertainty and futures images average all 64 samples, so the neighbour appears both standing and fallen). (b) The same 64 futures in the world model's latent space, on the real futures' principal axes.

Table 1: Fork probe with MuJoCo and real perception on unseen test sets: κ against the fork label and fork recall.

outcomes read from object positions	Set A (153 states) κ	Set A recall	Set B (160) κ	Set B recall	Set C (156) κ	Set C recall
ceiling (two oracle runs)	0.99	1.00	0.99	1.00	0.99	0.98
perception uncertainty ignored	0.42	0.65	0.54	0.65	0.42	0.67
fork probe	0.76	0.90	0.71	0.77	0.80	0.91
fork probe, 5 hypotheses	0.70	0.90	0.76	0.90	0.77	0.89
outcomes read from images: fork probe	0.75	0.88	0.71	0.77	0.78	0.88
5 hypotheses	–	–	–	–	**0.81**	**0.88**

(the split test on this latent agrees with the position-based test at κ 0.93 on held-out real futures) and predicts the next latent from the current one and the action with a flow-matching transition, drawing fresh noise at every step. It is trained on 8 futures per image whose start scenes were sampled within the measured perception error, so its target spread is the robot's own uncertainty; a deterministic version of the same network is the control. Both beat a no-motion baseline on 19-step latent error by 3–4×, yet on test set C the stochastic model reaches κ 0.19 and catches 18% of forks, and the deterministic one 0.52 and 60%, against 0.81 for the engine (Table 2).

Where the uncertainty went wrong. Table 2 compares with real futures the total spread of the 64 endings, its share on the main axis, and its ratio on fork versus other states. The stochastic model has the right *amount* of uncertainty but not its *shape* or *placement*: the spread is isotropic and nearly the same on forks and non-forks, the average uncertainty of futures rather than where they branch (Fig. 1b); other samplers are worse. Per-step noise is not wrong in general (diffusion weather ensembles stay well dispersed [11]); here it was misplaced. The deterministic model's tiny spread is aligned with the branch axis and 32× larger on forks, suggesting, as in the IPE, that uncertainty

Table 2: World models on test set C, and diagnostics on 60 held-out states with 64 real futures each.

rollouts from	κ	recall	spread	main axis	fork ratio
reference (MuJoCo; real futures)	0.81	0.88	1.00	0.57	4.8
token world model, stochastic	0.19	0.18	0.98	0.15	1.3
token world model, deterministic	0.52	0.60	0.04	0.69	32
deterministic, uncertain starts (held-out only)	0.29[†]	–	0.28	0.55	–

[†] agreement with real-future splits on held-out states; it failed this pre-test, so set C was not run.

should enter through the initial conditions. We tested this in a second, pre-registered iteration: a learned operator perturbs the start latent by sampled physical perception error, and deterministic dynamics are trained on full rollouts from each future's own start. It fails its held-out pre-test (κ 0.29), and even with exactly rendered perturbed starts it reaches only 0.19: the learned dynamics amplify start uncertainty too weakly (spread 0.28×) and form two groups on only half the real forks. The bottleneck is the dynamics, not the uncertainty input.

4 Related work

Simulation under uncertainty. Beyond the IPE and counterfactual accounts of physical support [12], robotics runs Monte Carlo rollouts over pose, friction or model parameters to score grasps and pushes [13, 14] or to build safety filters [15, 16]. Cannizzaro et al. [17] sample measured perception and placement noise through a simulator to predict whether a block tower stands, and Goyal [18] perturbs action chunks to measure, offline, how fast deviations grow. These score a task predicate, labels [19] or a growth rate; Zapolsky and Drumwright [20] proposed goal-free detection of divergent particle traces but evaluated task failure. We judge only whether the outcome set splits, a Monte Carlo test of proximity to a basin boundary [21].

Uncertainty-aware monitors. Latent safety filters use learned failure classifiers and epistemic uncertainty, also on Jenga [22, 5]; failure detectors use policy uncertainty calibrated on successes [6, 23]. These measure what a model does not know, not whether the physics is at a branch.

World models and their uncertainty. Learned simulators and latent world models [24, 25, 26, 27, 28] struggle with stiff contact [29], and audits find poor physical grounding and optimism under failure-inducing actions [30, 31, 32]. Yu et al. [33] propose evaluating world models on interventional action branches by per-branch error; our models have low error yet lose the forks. Stochastic and ensemble models [34, 35] are judged by likelihood or return, not by where their spread falls.

5 Conclusion and limitations

A robot can judge fragility the way people do, by propagating its own measured uncertainty through a rough simulator. For safe deployment, our results suggest three things. Physical uncertainty must be propagated, not just estimated. Doing so without a simulator needs a world model with branching fidelity, which our learned models lack despite low average error, so benchmarks should measure it. And until learned dynamics have it, the practical design is the IPE's: learned perception and measured uncertainty feeding an approximate physics engine.

Limitations. Simulation, one task, three tracked blocks; privileged MuJoCo geometry; one perception error model for all images; test sets vary; single training runs; chunks that fail every time are not flagged.

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